



FUTURE OF AI IN HEALTH

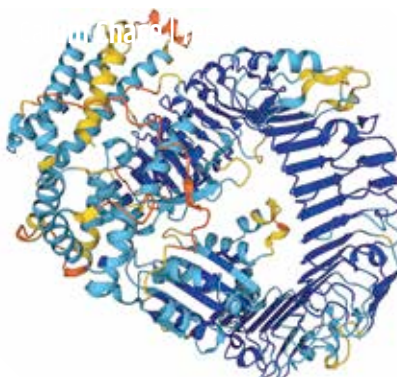
Getting medication from an AI. Is this the future?

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One of the biggest proofs that AI isn't just a passing fad, of AI's unparalleled transformative power, came in the form of DeepMind's AlphaFold – an AI program that effectively solved arguably the most difficult challenges in understanding one of the most fundamental building blocks of human biology and, in effect, all of life.

Before AlphaFold, modern science knew 3D structures for under 20% of roughly 20,000 proteins in the human body. These protein structures had been verified thanks to decades of tireless scientific pursuit, involving experimental methods like X-ray crystallography and nuclear magnetic resonance – needless to say, all of these cutting-edge methods to predict the 3D structure of a protein molecule wasn't just extremely expensive, it was also extremely time consuming. But thanks to DeepMind's AlphaFold, within just a year, 3D structures of 98.5% of all protein structures in the human body have been predicted – over 58% of them with high accuracy no less.

This is but just a tiny glimpse of what truly groundbreaking AI application can help humanity expand its knowledge of health sciences in a



AlphaFold

way that just wasn't possible until the AlphaFold breakthrough.

NEED FOR AI IN HEALTHCARE

Just think about your own experience of visiting a hospital or medical care facility in the recent past. From the point of booking a doctor's appointment to visiting the hospital, think of all the various touch points where technology can come in and make positive disruptions. What if there was a better, faster way of filling up the general admission form? Could technology step in to reduce all the administrative work done by nurses and doctors, day in and day out, so they have more quality time to spend with actual patient care?

There are a number of administrative applications for artificial intelligence in healthcare. The use of artificial intelligence in hospital settings is somewhat less game changing in this area as compared to patient care. But artificial intelligence in hospital administrative areas can provide substantial efficiencies. AI in healthcare can be used for a variety of applications, including claims processing, clinical documentation, revenue cycle management and medical records management.

One of the biggest challenges of infusing AI in healthcare at scale is the unavailability of data standardisation models – because artificial intelligence in healthcare requires access to large amounts of data in order to be effective, this is the most important problem that needs to be fixed by the healthcare industry as a whole. Another challenge is that there is a risk of bias if the data used to train the algorithms is not representative of the population as a whole. Finally, there is a lack of standardisation across different artificial intelligence systems, which can make it difficult to compare results or combine data from multiple sources.

EXPERT VIEW

Microsoft Research's Chief Medical